

Country Emissions Predictor

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Goals and Data

- Predict the carbon footprint of a country using projected energy consumption and renewable energy percentage
- Global sustainability dataset spanning 20 years
 - Per country each year
 - Source: <https://www.kaggle.com/datasets/anshtanwar/global-data-on-sustainable-energy>
- NASA GISTEMP v4
 - Global-mean monthly, seasonal, and annual means
 - Source: https://data.giss.nasa.gov/gistemp/tabledata_v4/GLB.Ts+dSST.csv

Steps and Services Used

- Steps
 - Upload datasets into cloud project bucket
 - Transform using BigQuery to create native tables
 - Join processed tables together into one
 - Connect to Looker Studio
- Services:
 - Google Cloud Storage
 - BigQuery
 - BigQuery ML
 - Looker Studio

Demo

Insights

- CO2 emissions are closely related to fossil fuel consumption
 - Transition to renewable energy is not aggressive enough
 - Fossil fuel is not being retired at an appropriate rate
- Highly developed countries have better carbon intensity
 - Gap in technology

Conclusion

- It is possible to predict emissions with high accuracy
 - Allowing what-if scenarios to be explored
- New demands like data centers are important variables to consider
 - Potential spikes from consumption